Homework, Lecture 5: Tacit collusion in oligopoly

Two firms, $P = 100 - q_1 - q_2$, $C(q_i) = 10q_i$, repeated forever with discount factor δ. Payoffs are normalised by $(1-\delta)$ (slide 21).

1. Stage game. $BR(q_j) = (90 - q_j)/2$. Cournot-Nash: $q_N = 30$, price 40, $\pi_N = 900$. Monopoly: $Q^m = 45$, price 55, profit 2025. Cartel: $q_C = 22.5$ each, $\pi_C = 1012.5$.

2. Best deviation. Against $q_C = 22.5$: $q_D = (90 - 22.5)/2 = 33.75$, price 43.75, $\pi_D = 33.75^2 = 1139.0625$. So $\pi_D > \pi_C > \pi_N$.

3. Grim trigger. $s_i(h) = 22.5$ if $h = \varnothing$ or both always produced 22.5; $s_i(h) = 30$ otherwise. One-stage deviation principle, two states. *Collusion state:*

$$\pi_C \geq (1-\delta)\pi_D + \delta\pi_N \iff \delta \geq \frac{\pi_D - \pi_C}{\pi_D - \pi_N} = \frac{126.5625}{239.0625} = \frac{9}{17} \approx 0.529.$$

Punishment state (off path): play is 30 forever and 30 is a best response to 30 (stage-game Nash equilibrium), so no one-shot deviation pays, for any δ. Hence grim trigger is an SPE iff $\delta \geq 9/17$. (For example, at $\delta = 1/2$ compliance gives 1012.5 against 1019.53125 from cheating; at $\delta = 0.6$ compliance wins.)

4. General case. $P = a - bQ$, cost $c < a$, $k = (a-c)^2/b$: $q_N = (a-c)/3b$, $\pi_N = k/9$; $q_C = (a-c)/4b$, $\pi_C = k/8$; $q_D = 3(a-c)/8b$, $\pi_D = 9k/64$. Then $\pi_D - \pi_C = k/64$ and $\pi_D - \pi_N = 17k/576$, so $\delta^* = 576/(64 \cdot 17) = 9/17$. The parameters only scale all profits by k, which cancels in the ratio.

5. What makes collusion easier. Collusion holds iff $(1-\delta)(\pi_D - \pi_C) \leq \delta(\pi_C - \pi_N)$: gain from cheating today against loss from punishment tomorrow.

- *Few firms.* With n firms, $\delta^*(n) = (n+1)^2/(n^2+6n+1)$: 9/17, 4/7, 25/41, 9/14 for $n = 2, \ldots, 5$ and 121/161 at $n = 10$; increasing in n, tending to 1.
- *Frequent interaction, patience.* $\delta = e^{-r\Delta t}$; the duopoly needs $r\Delta t \leq \ln(17/9) \approx 0.636$.
- *Observability, fast detection.* A long detection lag raises the gain from cheating; noisy signals force price wars (Green–Porter) and lower the value of collusion.
- *Growing demand* acts like a higher δ; declining demand, a known last period, or booms make cheating more tempting.
- *Symmetry*, so a single threshold binds; *multimarket contact*, since a deviation is punished in all markets; *capacity* matched to the cartel output.

A less harsh or shorter punishment (Abreu's carrot and stick) can lower the threshold below 9/17.

Sources: Lecture 5 slides 21, 25, 28–29, 34; Tadelis (2013) §10.4, p. 201; Cabral (2017) Ch. 9; Gibbons (1992) §2.3.C, p. 102. Numbers checked by `homework-lecture-05-check.py`.